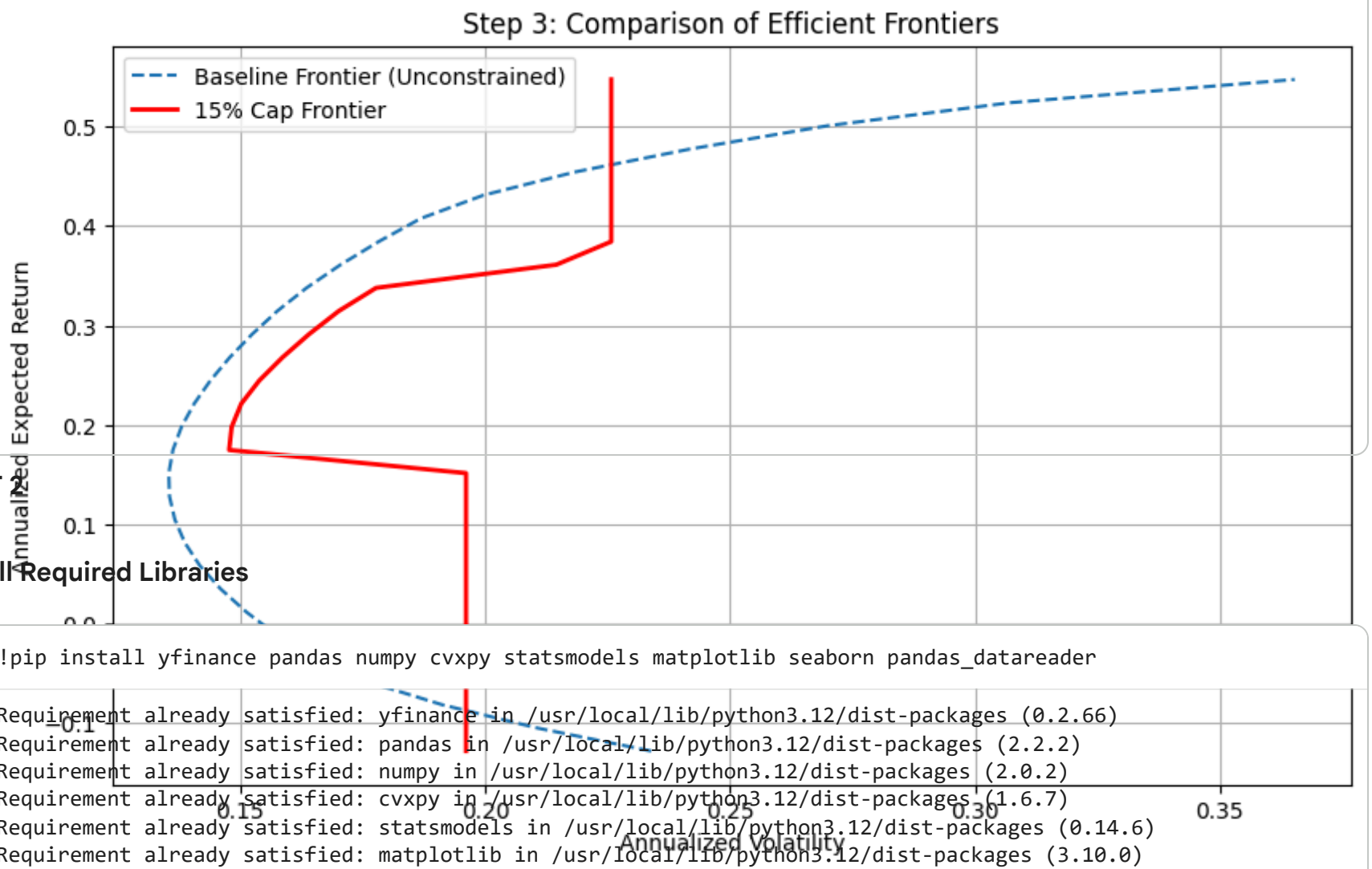
PART 1

```
import yfinance as yf
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.optimize import minimize


tickers = ['TSLA', 'WMT', 'BAC', 'GS', 'LLY', 'MRK', 'GOOG', 'META', 'AAPL', 'XOM']
start_date = '2023-07-01'
end_date = '2025-06-30'
oos_start = '2025-07-01'
oos_end = '2025-09-30'


df = yf.download(tickers, start=start_date, end=oos_end, auto_adjust=True)


if 'Close' in df.columns:
    data = df['Close']
else:
    data = df

train_data = data.loc[start_date:end_date]
oos_data = data.loc[oos_start:oos_end]


returns = train_data.pct_change().dropna()
mean_returns = returns.mean() * 252
cov_matrix = returns.cov() * 252


def portfolio_performance(weights, mean_returns, cov_matrix):
    p_ret = np.dot(weights, mean_returns)
```

```
    p_std = np.sqrt(np.dot(weights.T, np.dot(cov_matrix, weights)))
    return p_std, p_ret

def neg_sharpe_ratio(weights, mean_returns, cov_matrix, risk_free_rate=0.04):
    p_std, p_ret = portfolio_performance(weights, mean_returns, cov_matrix)
    return -(p_ret - risk_free_rate) / p_std

def solve_optimal_weights(mean_returns, cov_matrix, max_weight=1.0):
    num_assets = len(mean_returns)
    args = (mean_returns, cov_matrix)
    constraints = ({'type': 'eq', 'fun': lambda x: np.sum(x) - 1})
    bounds = tuple((0, max_weight) for _ in range(num_assets))


    init_guess = num_assets * [1./num_assets]

    result = minimize(neg_sharpe_ratio, init_guess,
                      args=args, method='SLSQP', bounds=bounds, constraints=constraints)
    return result.x


weights_step1 = solve_optimal_weights(mean_returns, cov_matrix, max_weight=1.0)


weights_step2 = solve_optimal_weights(mean_returns, cov_matrix, max_weight=0.15)


def get_frontier(mean_returns, cov_matrix, max_weight):
    # Create a range of target returns between the minimum and maximum mean returns
    target_returns = np.linspace(mean_returns.min(), mean_returns.max(), 30)
    frontier_volatility = []

    for target in target_returns:
        cons = ({'type': 'eq', 'fun': lambda x: np.sum(x) - 1},
                {'type': 'eq', 'fun': lambda x: np.dot(x, mean_returns) - target})
        bounds = tuple((0, max_weight) for _ in range(len(mean_returns)))
        res = minimize(lambda x: np.sqrt(np.dot(x.T, np.dot(cov_matrix, x))),
                       len(mean_returns) * [1./len(mean_returns)],
                       method='SLSQP', bounds=bounds, constraints=cons)
```

```
        frontier_volatility.append(res.fun)
    return frontier_volatility, target_returns

vol1, ret1 = get_frontier(mean_returns, cov_matrix, 1.0)
vol2, ret2 = get_frontier(mean_returns, cov_matrix, 0.15)

plt.figure(figsize=(10,6))
plt.plot(vol1, ret1, label='Baseline Frontier (Unconstrained)', linestyle='--')
plt.plot(vol2, ret2, label='15% Cap Frontier', color='red', linewidth=2)
plt.title('Step 3: Comparison of Efficient Frontiers')
plt.xlabel('Annualized Volatility')
plt.ylabel('Annualized Expected Return')
plt.grid(True)
plt.legend()
plt.show()

oos_returns = oos_data.pct_change().dropna()
port1_daily_rets = (oos_returns * weights_step1).sum(axis=1)
port2_daily_rets = (oos_returns * weights_step2).sum(axis=1)

print("-" * 30)
print("Step 1 Weights (Unconstrained):")
for t, w in zip(tickers, weights_step1):
    print(f"{t}: {w:.2%}")

print("\nStep 2 Weights (15% Cap):")
for t, w in zip(tickers, weights_step2):
    print(f"{t}: {w:.2%}")

print("-" * 30)
print(f"Step 1 OOS Cumulative Return: {((1 + port1_daily_rets).prod() - 1):.2%}")
print(f"Step 2 OOS Cumulative Return: {((1 + port2_daily_rets).prod() - 1):.2%}")
```

```
[*********************100%***********************]  10 of 10 completed
```



PART 2

Install Required Libraries

```
!pip install yfinance pandas numpy cvxpy statsmodels matplotlib seaborn pandas_datareader
```

```
Requirement already satisfied: yfinance in /usr/local/lib/python3.12/dist-packages (0.2.66)
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Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas)
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Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1
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Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4>=4
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Requirement already satisfied: certifi>=2024.2.2 in /usr/local/lib/python3.12/dist-packages (from curl_cffi>=0.
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Requirement already satisfied: pycparser in /usr/local/lib/python3.12/dist-packages (from cffi->clarabel>=0.5.0
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->osqp>=0
```

1. Import Libraries

```
import numpy as np
import pandas as pd
import yfinance as yf
import cvxpy as cp
import statsmodels.api as sm
```

```
import matplotlib.pyplot as plt
import seaborn as sns
from pandas_datareader import data as web
```

2. Download Stock Price Data (July 2023 – June 2025)

```
tickers = ['TSLA','WMT','BAC','GS','LLY','MRK','GOOG','META','AAPL','XOM']

start_date = "2023-07-01"
end_date = "2025-06-30"

prices = yf.download(
    tickers,
    start=start_date,
    end=end_date,
    auto_adjust=True
)

returns = prices.pct_change().dropna()
returns.head()
```

```
[*********************100%***********************]  10 of 10 completed
```

Price	Close										..
Ticker	AAPL	BAC	GOOG	GS	LLY	META	MRK	TSLA	WMT	XOM	..
Date											
2023-07-05	-0.005871	-0.004110	0.017170	-0.020085	0.009276	0.029194	-0.005510	0.009506	-0.000569	-0.005118	.
2023-07-06	0.002509	-0.027510	-0.013863	-0.022028	-0.005347	-0.008085	-0.017151	-0.021028	-0.006325	-0.037321	.
2023-07-07	-0.005891	0.008840	-0.006533	0.006933	-0.024028	-0.005000	-0.024608	-0.007630	-0.023041	0.002332	.
2023-07-10	-0.010856	0.004557	-0.027218	0.004125	0.002389	0.012288	0.008807	-0.017564	0.009056	0.014831	.
2023-07-11	-0.002810	0.012561	0.007187	0.013935	-0.030387	0.014247	-0.011459	0.000668	-0.001485	0.012227	.

5 rows × 50 columns

3. Compute Expected Returns & Covariance Matrix

```
mu = returns.mean()
Sigma = returns.cov()

mu, Sigma
```

```
(Price   Ticker
 Close   AAPL      0.000257
         BAC       0.001206
         GOOG      0.000974
         GS        0.001767
         LLY       0.001310
         META      0.002164
         MRK      -0.000510
```

```
            TSLA      0.001081
            WMT       0.001365
            XOM       0.000275
      High  AAPL      0.000227
            BAC       0.001203
            GOOG      0.000947
            GS        0.001732
            LLY       0.001341
            META      0.002128
            MRK      -0.000530
            TSLA      0.000879
            WMT       0.001353
            XOM       0.000252
      Low   AAPL      0.000242
            BAC       0.001230
            GOOG      0.000915
            GS        0.001755
            LLY       0.001330
            META      0.002129
            MRK      -0.000529
            TSLA      0.000934
            WMT       0.001356
            XOM       0.000247
      Open  AAPL      0.000278
            BAC       0.001260
            GOOG      0.000951
            GS        0.001801
            LLY       0.001360
            META      0.002204
            MRK      -0.000510
            TSLA      0.001148
            WMT       0.001366
            XOM       0.000302
      Volume AAPL     0.058109
            BAC       0.054664
            GOOG      0.072575
            GS        0.070105
            LLY       0.094397
            META      0.072834
            MRK       0.090217
            TSLA      0.030576
            WMT       0.074048
```

```
         XOM       0.051001
dtype: float64,
Price             Close                                                \
Ticker             AAPL       BAC          GOOG        GS       LLY
Price  Ticker
Close  AAPL    0.000301  0.000098  1.485862e-04  0.000130  0.000083
       BAC     0.000098  0.000273  7.802768e-05  0.000233  0.000044
```

4. Mean–Variance Optimization (Max Weight = 15%)

```
close_returns = returns["Close"]
```

```
mu_close = close_returns.mean()
Sigma_close = close_returns.cov()
```

```
import cvxpy as cp
import pandas as pd

n = len(mu_close)
w = cp.Variable(n)

objective = cp.Minimize(cp.quad_form(w, Sigma_close))

constraints = [
    cp.sum(w) == 1,
    w >= 0,
    w <= 0.15
]

problem = cp.Problem(objective, constraints)
problem.solve()

optimal_weights = pd.Series(w.value, index=mu_close.index)

optimal_weights.sort_values(ascending=False)
```

Ticker	0
BAC	1.500000e-01
GOOG	1.500000e-01
WMT	1.500000e-01
XOM	1.500000e-01
MRK	1.500000e-01
AAPL	1.193168e-01
LLY	1.186119e-01
GS	1.123714e-02
META	8.341653e-04
TSLA	-4.495229e-20

dtype: float64

5. Compute Portfolio Returns

```
close_returns = returns["Close"]
```

```
close_returns = close_returns[optimal_weights.index]
```

```
portfolio_returns = close_returns.dot(optimal_weights)
portfolio_returns.head()

print(close_returns.columns.tolist())
print(optimal_weights.index.tolist())
```

```
['AAPL', 'BAC', 'GOOG', 'GS', 'LLY', 'META', 'MRK', 'TSLA', 'WMT', 'XOM']
['AAPL', 'BAC', 'GOOG', 'GS', 'LLY', 'META', 'MRK', 'TSLA', 'WMT', 'XOM']
```

6. Download Fama–French 5-Factor Data

```
ff_factors = web.DataReader('F-F_Research_Data_5_Factors_2x3_daily',
                            'famafrench', start=start_date)[0]

ff_factors.index = pd.to_datetime(ff_factors.index)

ff_factors = ff_factors / 100
ff_factors.head()
```

```
/tmp/ipython-input-825446906.py:1: FutureWarning: The argument 'date_parser' is deprecated and will be removed
  ff_factors = web.DataReader('F-F_Research_Data_5_Factors_2x3_daily',
```

Date	Mkt-RF	SMB	HML	RMW	CMA	RF
2023-07-03	0.0015	0.0027	0.0106	0.0018	0.0011	0.0002
2023-07-05	-0.0029	-0.0106	-0.0026	-0.0030	-0.0001	0.0002
2023-07-06	-0.0094	-0.0058	-0.0069	0.0057	0.0005	0.0002
2023-07-07	-0.0013	0.0130	0.0121	-0.0023	0.0018	0.0002
2023-07-10	0.0036	0.0124	-0.0001	-0.0103	0.0054	0.0002

7. Align Portfolio & Factor Data

```
data = pd.concat([portfolio_returns, ff_factors], axis=1).dropna()

data.columns = ['Portfolio'] + list(ff_factors.columns)

data.head()
```

Date	Portfolio	Mkt-RF	SMB	HML	RMW	CMA	RF
2023-07-05	0.000478	-0.0029	-0.0106	-0.0026	-0.0030	-0.0001	0.0002
2023-07-06	-0.015915	-0.0094	-0.0058	-0.0069	0.0057	0.0005	0.0002
2023-07-07	-0.009931	-0.0013	0.0130	0.0121	-0.0023	0.0018	0.0002
2023-07-10	0.000550	0.0036	0.0124	-0.0001	-0.0103	0.0054	0.0002
2023-07-11	-0.000916	0.0076	0.0018	0.0048	-0.0021	0.0028	0.0002

8. Compute Excess Portfolio Returns

```
data['Excess_Portfolio'] = data['Portfolio'] - data['RF']
```

9. OLS Regression (Fama–French 5-Factor Model)

```
X = data[['Mkt-RF','SMB','HML','RMW','CMA']]
X = sm.add_constant(X)
Y = data['Excess_Portfolio']

ols_model = sm.OLS(Y, X).fit()
print(ols_model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:       Excess_Portfolio   R-squared:                       0.708
Model:                            OLS   Adj. R-squared:                  0.705
```

```
Method:                 Least Squares   F-statistic:                     238.5
Date:                Tue, 03 Feb 2026   Prob (F-statistic):          5.71e-129
Time:                        16:40:44   Log-Likelihood:                 1929.1
No. Observations:                 498   AIC:                            -3846.
Df Residuals:                     492   BIC:                            -3821.
Df Model:                           5
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const       -1.88e-05      0.000     -0.082      0.934      -0.000       0.000
Mkt-RF         0.8019      0.024     32.886      0.000       0.754       0.850
SMB           -0.1056      0.039     -2.734      0.006      -0.181      -0.030
HML            0.2047      0.039      5.256      0.000       0.128       0.281
RMW            0.1074      0.053      2.035      0.042       0.004       0.211
CMA           -0.0373      0.053     -0.703      0.483      -0.142       0.067
==============================================================================
Omnibus:                       27.671   Durbin-Watson:                   1.967
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               72.014
Skew:                           0.217   Prob(JB):                     2.30e-16
Kurtosis:                       4.812   Cond. No.                         271.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

10. Robust Regression (HC3 Standard Errors)

```
robust_model = sm.OLS(Y, X).fit(cov_type='HC3')
print(robust_model.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:       Excess_Portfolio   R-squared:                       0.708
Model:                            OLS   Adj. R-squared:                  0.705
Method:                 Least Squares   F-statistic:                     125.5
Date:                Tue, 03 Feb 2026   Prob (F-statistic):           1.80e-85
Time:                        16:40:48   Log-Likelihood:                 1929.1
No. Observations:                 498   AIC:                            -3846.
Df Residuals:                     492   BIC:                            -3821.
```

```
Df Model:                           5
Covariance Type:                  HC3
==============================================================================
                 coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------
const       -1.88e-05      0.000     -0.081      0.935      -0.000       0.000
Mkt-RF         0.8019      0.035     22.956      0.000       0.733       0.870
SMB           -0.1056      0.048     -2.204      0.028      -0.200      -0.012
HML            0.2047      0.056      3.648      0.000       0.095       0.315
RMW            0.1074      0.066      1.615      0.106      -0.023       0.238
CMA           -0.0373      0.078     -0.480      0.631      -0.190       0.115
==============================================================================
Omnibus:                       27.671   Durbin-Watson:                   1.967
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               72.014
Skew:                           0.217   Prob(JB):                     2.30e-16
Kurtosis:                       4.812   Cond. No.                         271.
==============================================================================

Notes:
[1] Standard Errors are heteroscedasticity robust (HC3)
```

11. Training / Testing Split (Time-Series Safe)

```
split_date = "2025-01-01"

train = data.loc[:split_date]
test = data.loc[split_date:]

X_train = sm.add_constant(train[['Mkt-RF','SMB','HML','RMW','CMA']])
Y_train = train['Excess_Portfolio']

X_test = sm.add_constant(test[['Mkt-RF','SMB','HML','RMW','CMA']])
Y_test = test['Excess_Portfolio']

train_model = sm.OLS(Y_train, X_train).fit()

predictions = train_model.predict(X_test)
```

12. Out-of-Sample Performance Metrics

```
rmse = np.sqrt(((Y_test - predictions)**2).mean())
r2_out = 1 - ((Y_test - predictions)**2).sum() / ((Y_test - Y_test.mean())**2).sum()

rmse, r2_out
```

```
(np.float64(0.006758878938928171), np.float64(0.7767238003005621))
```

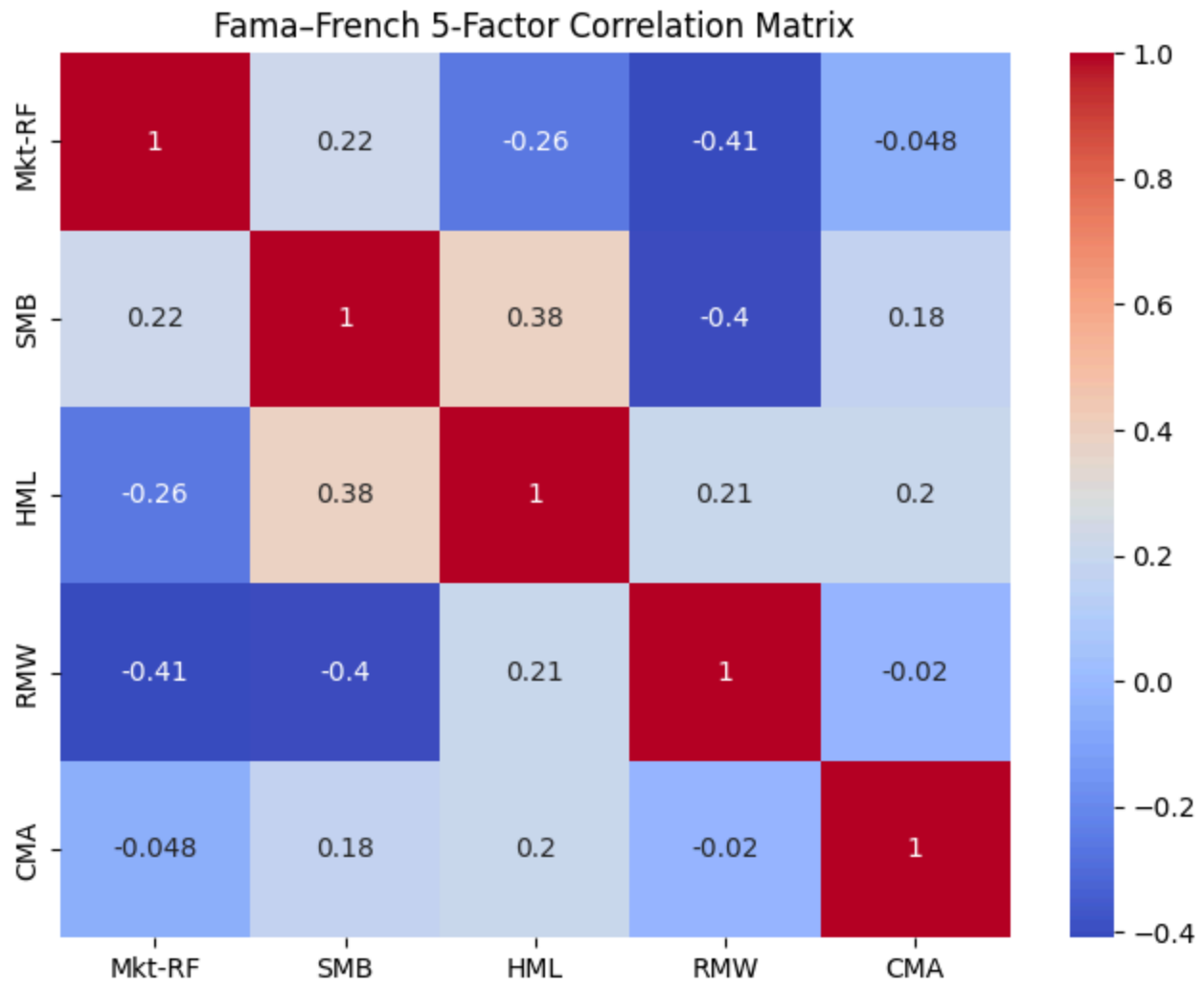
13. Correlation Matrix of Fama–French Factors

```
corr_matrix = ff_factors[['Mkt-RF','SMB','HML','RMW','CMA']].corr()
corr_matrix
```

	Mkt-RF	SMB	HML	RMW	CMA
Mkt-RF	1.000000	0.216287	-0.257876	-0.408714	-0.048203
SMB	0.216287	1.000000	0.378243	-0.400338	0.175107
HML	-0.257876	0.378243	1.000000	0.209535	0.202981
RMW	-0.408714	-0.400338	0.209535	1.000000	-0.019523
CMA	-0.048203	0.175107	0.202981	-0.019523	1.000000

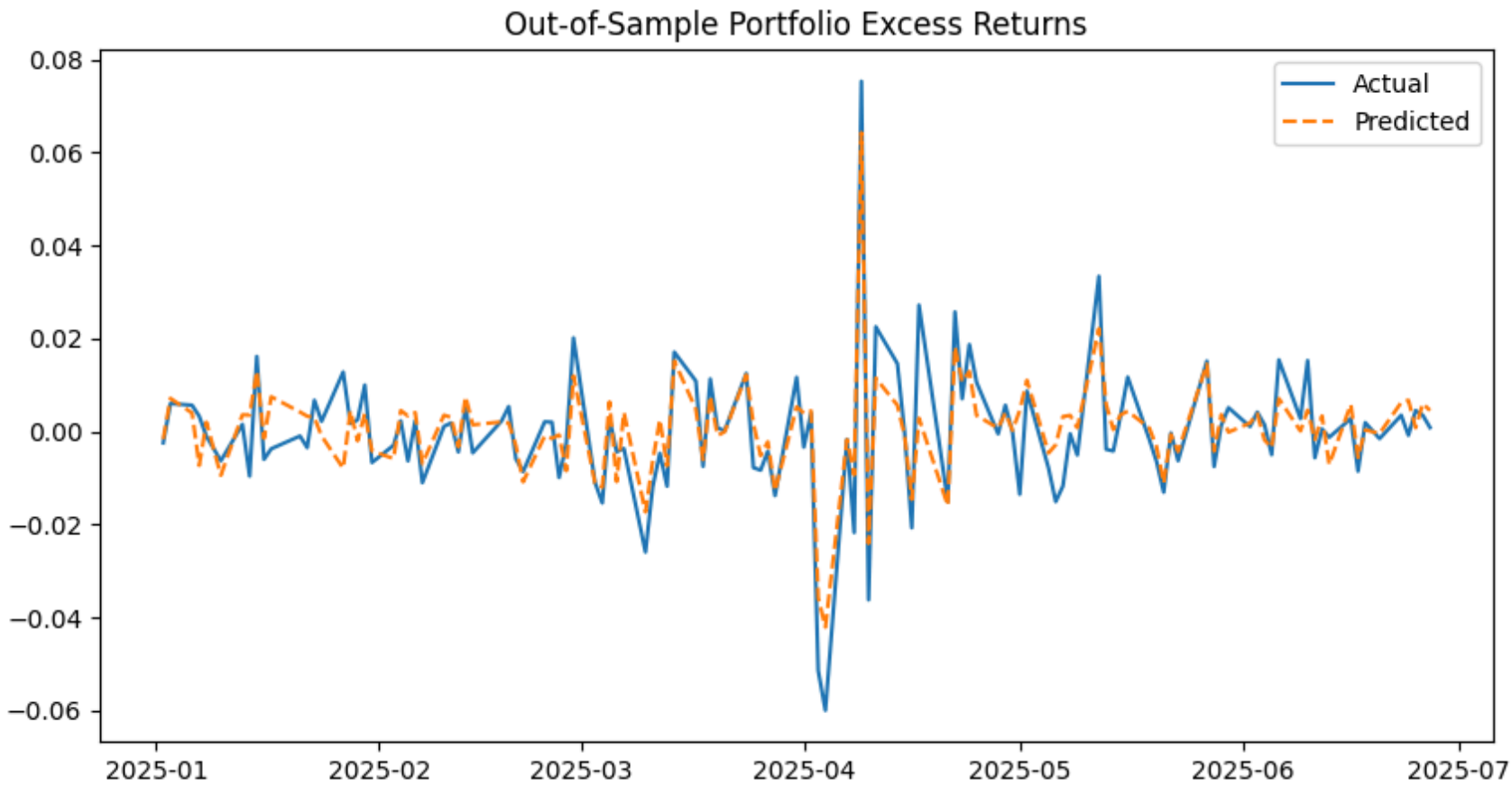
14. Heatmap Visualization

```
plt.figure(figsize=(8,6))
sns.heatmap(corr_matrix, annot=True, cmap="coolwarm")
plt.title("Fama–French 5-Factor Correlation Matrix")
plt.show()
```



15. Plot Actual vs Predicted Returns

```
plt.figure(figsize=(10,5))
plt.plot(Y_test.index, Y_test, label="Actual")
plt.plot(predictions.index, predictions, label="Predicted", linestyle="--")
plt.title("Out-of-Sample Portfolio Excess Returns")
plt.legend()
plt.show()
```



16. Export Tables to CSV

```
optimal_weights.to_csv("optimal_weights.csv")
corr_matrix.to_csv("factor_correlation_matrix.csv")
```

PART 3.

```
import numpy as np
import pandas as pd
from datetime import datetime
from scipy.optimize import minimize
import warnings
warnings.filterwarnings('ignore')
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_style("whitegrid")
plt.rcParams['figure.figsize'] = (14, 8)

asset_universe = ['TSLA', 'WMT', 'BAC', 'LLY', 'AAPL']
num_assets = len(asset_universe)

period_training_start = datetime(2023, 7, 1)
period_training_end = datetime(2025, 6, 30)

trading_dates_range = pd.bdate_range(start=period_training_start, end=period_training_end)
trading_days_count = len(trading_dates_range)

asset_annual_returns = np.array([0.28, 0.08, 0.12, 0.35, 0.18])
asset_annual_volatilities = np.array([0.65, 0.18, 0.25, 0.48, 0.28])

correlation_matrix = np.array([
    [1.00, 0.15, 0.22, 0.35, 0.40],
    [0.15, 1.00, 0.35, 0.08, 0.25],
    [0.22, 0.35, 1.00, 0.18, 0.32],
    [0.35, 0.08, 0.18, 1.00, 0.28],
    [0.40, 0.25, 0.32, 0.28, 1.00]
])

covariance_matrix_daily = np.outer(asset_annual_volatilities, asset_annual_volatilities) * correlation_matrix

np.random.seed(42)
```

```
daily_log_returns = np.random.multivariate_normal(
    mean=asset_annual_returns / 252,
    cov=covariance_matrix_daily,
    size=trading_days_count
)

initial_price_vector = np.array([200.0, 80.0, 30.0, 600.0, 180.0])
cumulative_return_factors = np.exp(np.cumsum(daily_log_returns, axis=0))
synthetic_price_matrix = initial_price_vector * cumulative_return_factors

price_dataframe = pd.DataFrame(synthetic_price_matrix, columns=asset_universe, index=trading_dates_range)
daily_returns = price_dataframe.pct_change().dropna()

annual_expected_returns = daily_returns.mean() * 252
covariance_matrix_annual = daily_returns.cov() * 252

print("="*80)
print("STEP 7: UNCONSTRAINED OPTIMIZATION VS SIMULATION")
print("="*80)
print("\nAsset Universe:", asset_universe)
print("\nTraining Period: July 1, 2023 - June 30, 2025")
print(f"\nData Points Available: {len(daily_returns)} trading days")
print("\nAnnualized Expected Returns:")
for asset_idx, ticker in enumerate(asset_universe):
    print(f"  {ticker}: {annual_expected_returns.iloc[asset_idx]:.6f} ({annual_expected_returns.iloc[asset_idx
print("\nAnnualized Covariance Matrix:")
print(covariance_matrix_annual)

def portfolio_performance_metrics(weight_allocation, expected_return_vector, cov_matrix_annual):
    portfolio_annual_return = np.dot(weight_allocation, expected_return_vector.values)
    portfolio_annual_volatility = np.sqrt(np.dot(weight_allocation, np.dot(cov_matrix_annual.values, weight_al
    portfolio_sharpe_ratio = portfolio_annual_return / portfolio_annual_volatility if portfolio_annual_volatil
    return portfolio_annual_return, portfolio_annual_volatility, portfolio_sharpe_ratio

def negative_sharpe_objective(weight_allocation, expected_return_vector, cov_matrix_annual):
    portfolio_return, portfolio_volatility, sharpe_metric = portfolio_performance_metrics(weight_allocation, e
    return -sharpe_metric if portfolio_volatility > 0 else 0

constraints_step7 = {'type': 'eq', 'fun': lambda weights: np.sum(weights) - 1.0}
```

```
bounds_step7 = tuple((0.0, 1.0) for _ in range(num_assets))
initial_weight_guess = np.array([1.0/num_assets] * num_assets)

optimization_solution_step7 = minimize(
    negative_sharpe_objective,
    initial_weight_guess,
    args=(annual_expected_returns, covariance_matrix_annual),
    method='SLSQP',
    bounds=bounds_step7,
    constraints=constraints_step7,
    options={'ftol': 1e-12, 'maxiter': 1000}
)

optimal_weights_step7 = optimization_solution_step7.x
optimal_return_step7, optimal_volatility_step7, optimal_sharpe_step7 = portfolio_performance_metrics(
    optimal_weights_step7, annual_expected_returns, covariance_matrix_annual
)

print("\n" + "="*80)
print("STEP 7: OPTIMIZATION APPROACH")
print("="*80)
print("\nOptimal Portfolio Weights (Optimization):")
for ticker_idx, ticker_name in enumerate(asset_universe):
    print(f"  {ticker_name}: {optimal_weights_step7[ticker_idx]:.6f} ({optimal_weights_step7[ticker_idx]*100:.
print(f"\nOptimal Portfolio Annual Return: {optimal_return_step7:.6f} ({optimal_return_step7*100:.2f}%)")
print(f"Optimal Portfolio Annual Volatility: {optimal_volatility_step7:.6f} ({optimal_volatility_step7*100:.2f
print(f"Optimal Sharpe Ratio: {optimal_sharpe_step7:.6f}")

simulation_sample_sizes_step7 = [10000, 50000, 100000, 250000, 500000, 1000000]
convergence_results_step7 = []

np.random.seed(42)

for sample_size in simulation_sample_sizes_step7:
    random_weights_matrix = np.random.dirichlet(np.ones(num_assets), sample_size)
    simulated_sharpe_values = []

    for random_portfolio_weights in random_weights_matrix:
        sim_port_return, sim_port_volatility, sim_port_sharpe = portfolio_performance_metrics(
```

```
                random_portfolio_weights, annual_expected_returns, covariance_matrix_annual
            )
            simulated_sharpe_values.append(sim_port_sharpe)

        best_sim_index = np.argmax(simulated_sharpe_values)
        best_simulated_weights = random_weights_matrix[best_sim_index]
        best_sim_return, best_sim_volatility, best_sim_sharpe = portfolio_performance_metrics(
            best_simulated_weights, annual_expected_returns, covariance_matrix_annual
        )

        weight_l1_distance = np.sum(np.abs(best_simulated_weights - optimal_weights_step7))
        sharpe_absolute_difference = np.abs(best_sim_sharpe - optimal_sharpe_step7)
        sharpe_percentage_deviation = (sharpe_absolute_difference / optimal_sharpe_step7) * 100 if optimal_sharpe_

        convergence_results_step7.append({
            'Simulations': sample_size,
            'Optimal_Weights': best_simulated_weights,
            'Portfolio_Return': best_sim_return,
            'Portfolio_Volatility': best_sim_volatility,
            'Sharpe_Ratio': best_sim_sharpe,
            'L1_Distance': weight_l1_distance,
            'Sharpe_Difference': sharpe_absolute_difference,
            'Sharpe_Error_Pct': sharpe_percentage_deviation
        })

        print(f"\nSimulation Count: {sample_size}")
        print(f"  Best Simulated Weights: {[f'{w:.6f}' for w in best_simulated_weights]}")
        print(f"  Best Simulated Sharpe Ratio: {best_sim_sharpe:.6f}")
        print(f"  L1 Distance from Optimal: {weight_l1_distance:.8f}")
        print(f"  Sharpe Ratio Difference: {sharpe_absolute_difference:.8f} ({sharpe_percentage_deviation:.4f}%)")

    convergence_table_step7 = pd.DataFrame([
        {
            'Sample_Size': r['Simulations'],
            'L1_Weight_Distance': r['L1_Distance'],
            'Sharpe_Error_Percentage': r['Sharpe_Error_Pct']
        }
        for r in convergence_results_step7
    ])
```

```
print("\n" + "-"*80)
print("STEP 7: CONVERGENCE ANALYSIS")
print("-"*80)
print(convergence_table_step7.to_string(index=False))

figure_step7, subplot_axes_step7 = plt.subplots(1, 2, figsize=(14, 5))

simulation_counts = [r['Simulations'] for r in convergence_results_step7]
weight_distances = [r['L1_Distance'] for r in convergence_results_step7]
sharpe_errors_percentage = [r['Sharpe_Error_Pct'] for r in convergence_results_step7]

subplot_axes_step7[0].plot(simulation_counts, weight_distances, marker='o', linewidth=2.5, markersize=9, color
subplot_axes_step7[0].set_xscale('log')
subplot_axes_step7[0].set_yscale('log')
subplot_axes_step7[0].set_xlabel('Number of Monte Carlo Simulations', fontsize=12, fontweight='bold')
subplot_axes_step7[0].set_ylabel('L1 Distance from Analytical Optimum', fontsize=12, fontweight='bold')
subplot_axes_step7[0].set_title('Step 7: Portfolio Weight Convergence Analysis\n(Unconstrained Optimization)',
subplot_axes_step7[0].grid(True, alpha=0.3, linestyle='--')
subplot_axes_step7[0].legend(fontsize=10)

subplot_axes_step7[1].plot(simulation_counts, sharpe_errors_percentage, marker='s', linewidth=2.5, markersize=
subplot_axes_step7[1].set_xscale('log')
subplot_axes_step7[1].set_yscale('log')
subplot_axes_step7[1].set_xlabel('Number of Monte Carlo Simulations', fontsize=12, fontweight='bold')
subplot_axes_step7[1].set_ylabel('Sharpe Ratio Error (%)', fontsize=12, fontweight='bold')
subplot_axes_step7[1].set_title('Step 7: Sharpe Ratio Convergence Analysis\n(Unconstrained Optimization)', fon
subplot_axes_step7[1].grid(True, alpha=0.3, linestyle='--')
subplot_axes_step7[1].legend(fontsize=10)

plt.tight_layout()
plt.show()

print("\n" + "="*80)
print("STEP 7: KEY FINDINGS")
print("="*80)
print(f"\nUnconstrained Portfolio Optimization Results:")
print(f"  - Number of actively invested assets: {sum(1 for w in optimal_weights_step7 if w > 0.05)}")
print(f"  - Optimal Sharpe Ratio (Analytical): {optimal_sharpe_step7:.6f}")
```

```
print(f"  - Optimal Annual Return: {optimal_return_step7*100:.2f}%")
print(f"  - Optimal Annual Volatility: {optimal_volatility_step7*100:.2f}%")

convergence_threshold_1pct = None
for result_entry in convergence_results_step7:
    if result_entry['Sharpe_Error_Pct'] <= 1.0:
        convergence_threshold_1pct = result_entry['Simulations']
        break

if convergence_threshold_1pct:
    print(f"\nSimulations Required for 1% Sharpe Ratio Error: {convergence_threshold_1pct:,}")
else:
    print(f"\nSimulations Required for 1% Sharpe Ratio Error: Not achieved in tested range")

print("\n" + "="*80)
print("STEP 8: BOX-CONSTRAINED OPTIMIZATION VS SIMULATION")
print("="*80)

maximum_weight_limit = 0.30
minimum_weight_limit = 0.0

constraints_step8 = [
    {'type': 'eq', 'fun': lambda weights: np.sum(weights) - 1.0},
]
bounds_step8 = tuple((minimum_weight_limit, maximum_weight_limit) for _ in range(num_assets))

optimization_solution_step8 = minimize(
    negative_sharpe_objective,
    initial_weight_guess,
    args=(annual_expected_returns, covariance_matrix_annual),
    method='SLSQP',
    bounds=bounds_step8,
    constraints=constraints_step8,
    options={'ftol': 1e-12, 'maxiter': 1000}
)

optimal_weights_step8 = optimization_solution_step8.x
optimal_return_step8, optimal_volatility_step8, optimal_sharpe_step8 = portfolio_performance_metrics(
    optimal_weights_step8, annual_expected_returns, covariance_matrix_annual
```

```
)

print("\nBox Constraint Applied: Maximum Weight per Asset = 30%")
print("\nOptimal Portfolio Weights (Optimization with Box Constraints):")
for ticker_idx, ticker_name in enumerate(asset_universe):
    print(f"  {ticker_name}: {optimal_weights_step8[ticker_idx]:.6f} ({optimal_weights_step8[ticker_idx]*100:.
print(f"\nOptimal Portfolio Annual Return: {optimal_return_step8:.6f} ({optimal_return_step8*100:.2f}%)")
print(f"Optimal Portfolio Annual Volatility: {optimal_volatility_step8:.6f} ({optimal_volatility_step8*100:.2f
print(f"Optimal Sharpe Ratio: {optimal_sharpe_step8:.6f}")

simulation_sample_sizes_step8 = [500000]
convergence_results_step8 = []

np.random.seed(42)

for sample_size in simulation_sample_sizes_step8:
    valid_portfolio_count = 0
    valid_weights_list = []
    valid_sharpe_list = []

    attempts = 0
    max_attempts = 50

    while valid_portfolio_count < sample_size and attempts < max_attempts:
        candidate_weights_matrix = np.random.dirichlet(np.ones(num_assets), sample_size * 3)

        for candidate_weight_vector in candidate_weights_matrix:
            if np.all(candidate_weight_vector <= maximum_weight_limit) and np.all(candidate_weight_vector >= m
                valid_weights_list.append(candidate_weight_vector)
                cand_return, cand_volatility, cand_sharpe = portfolio_performance_metrics(
                    candidate_weight_vector, annual_expected_returns, covariance_matrix_annual
                )
                valid_sharpe_list.append(cand_sharpe)
                valid_portfolio_count += 1

                if valid_portfolio_count >= sample_size:
                    break

        attempts += 1
```

```
        best_sim_index_s8 = np.argmax(valid_sharpe_list)
        best_simulated_weights_s8 = np.array(valid_weights_list[best_sim_index_s8])
        best_sim_return_s8, best_sim_volatility_s8, best_sim_sharpe_s8 = portfolio_performance_metrics(
            best_simulated_weights_s8, annual_expected_returns, covariance_matrix_annual
        )

        weight_l1_distance_s8 = np.sum(np.abs(best_simulated_weights_s8 - optimal_weights_step8))
        sharpe_absolute_difference_s8 = np.abs(best_sim_sharpe_s8 - optimal_sharpe_step8)
        sharpe_percentage_deviation_s8 = (sharpe_absolute_difference_s8 / optimal_sharpe_step8) * 100 if optimal_s

        convergence_results_step8.append({
            'Simulations': sample_size,
            'Optimal_Weights': best_simulated_weights_s8,
            'Portfolio_Return': best_sim_return_s8,
            'Portfolio_Volatility': best_sim_volatility_s8,
            'Sharpe_Ratio': best_sim_sharpe_s8,
            'L1_Distance': weight_l1_distance_s8,
            'Sharpe_Difference': sharpe_absolute_difference_s8,
            'Sharpe_Error_Pct': sharpe_percentage_deviation_s8
        })

        print(f"\nValid Simulations Generated: {sample_size}")
        print(f"  Best Simulated Weights: {[f'{w:.6f}' for w in best_simulated_weights_s8]}")
        print(f"  Best Simulated Sharpe Ratio: {best_sim_sharpe_s8:.6f}")
        print(f"  L1 Distance from Optimal: {weight_l1_distance_s8:.8f}")
        print(f"  Sharpe Ratio Difference: {sharpe_absolute_difference_s8:.8f} ({sharpe_percentage_deviation_s8:.4

    convergence_table_step8 = pd.DataFrame([
        {
            'Sample_Size': r['Simulations'],
            'L1_Weight_Distance': r['L1_Distance'],
            'Sharpe_Error_Percentage': r['Sharpe_Error_Pct']
        }
        for r in convergence_results_step8
    ])

    print("\n" + "-"*80)
    print("STEP 8: CONVERGENCE ANALYSIS (WITH BOX CONSTRAINTS)")
```

```
print("-"*80)
print(convergence_table_step8.to_string(index=False))

figure_step8, subplot_axes_step8 = plt.subplots(1, 2, figsize=(14, 5))

simulation_counts_s8 = [r['Simulations'] for r in convergence_results_step8]
weight_distances_s8 = [r['L1_Distance'] for r in convergence_results_step8]
sharpe_errors_percentage_s8 = [r['Sharpe_Error_Pct'] for r in convergence_results_step8]

subplot_axes_step8[0].scatter(simulation_counts_s8, weight_distances_s8, s=350, marker='D', color='darkgreen',
subplot_axes_step8[0].set_xscale('log')
subplot_axes_step8[0].set_yscale('log')
subplot_axes_step8[0].set_xlabel('Valid Constrained Portfolios Evaluated', fontsize=12, fontweight='bold')
subplot_axes_step8[0].set_ylabel('L1 Distance from Analytical Optimum', fontsize=12, fontweight='bold')
subplot_axes_step8[0].set_title('Step 8: Weight Convergence Validation\n(Box-Constrained: Max 30% per Asset)',
subplot_axes_step8[0].grid(True, alpha=0.3, linestyle='--')
subplot_axes_step8[0].legend(fontsize=10)

subplot_axes_step8[1].scatter(simulation_counts_s8, sharpe_errors_percentage_s8, s=350, marker='D', color='dar
subplot_axes_step8[1].set_xscale('log')
subplot_axes_step8[1].set_yscale('log')
subplot_axes_step8[1].set_xlabel('Valid Constrained Portfolios Evaluated', fontsize=12, fontweight='bold')
subplot_axes_step8[1].set_ylabel('Sharpe Ratio Error (%)', fontsize=12, fontweight='bold')
subplot_axes_step8[1].set_title('Step 8: Sharpe Error Validation\n(Box-Constrained: Max 30% per Asset)', fonts
subplot_axes_step8[1].grid(True, alpha=0.3, linestyle='--')
subplot_axes_step8[1].legend(fontsize=10)

plt.tight_layout()
plt.show()

figure_comparison, subplot_axes_comparison = plt.subplots(1, 2, figsize=(16, 6))

subplot_axes_comparison[0].plot(simulation_counts, sharpe_errors_percentage, marker='o', linewidth=3.5, marker
subplot_axes_comparison[0].scatter([500000], sharpe_errors_percentage_s8, s=450, marker='s', label='Step 8 Box
subplot_axes_comparison[0].set_xscale('log')
subplot_axes_comparison[0].set_yscale('log')
subplot_axes_comparison[0].set_xlabel('Number of Simulations / Valid Portfolios', fontsize=13, fontweight='bol
subplot_axes_comparison[0].set_ylabel('Sharpe Ratio Error (%)', fontsize=13, fontweight='bold')
subplot_axes_comparison[0].set_title('Convergence Comparison: Unconstrained vs Box-Constrained\nMonte Carlo Va
```

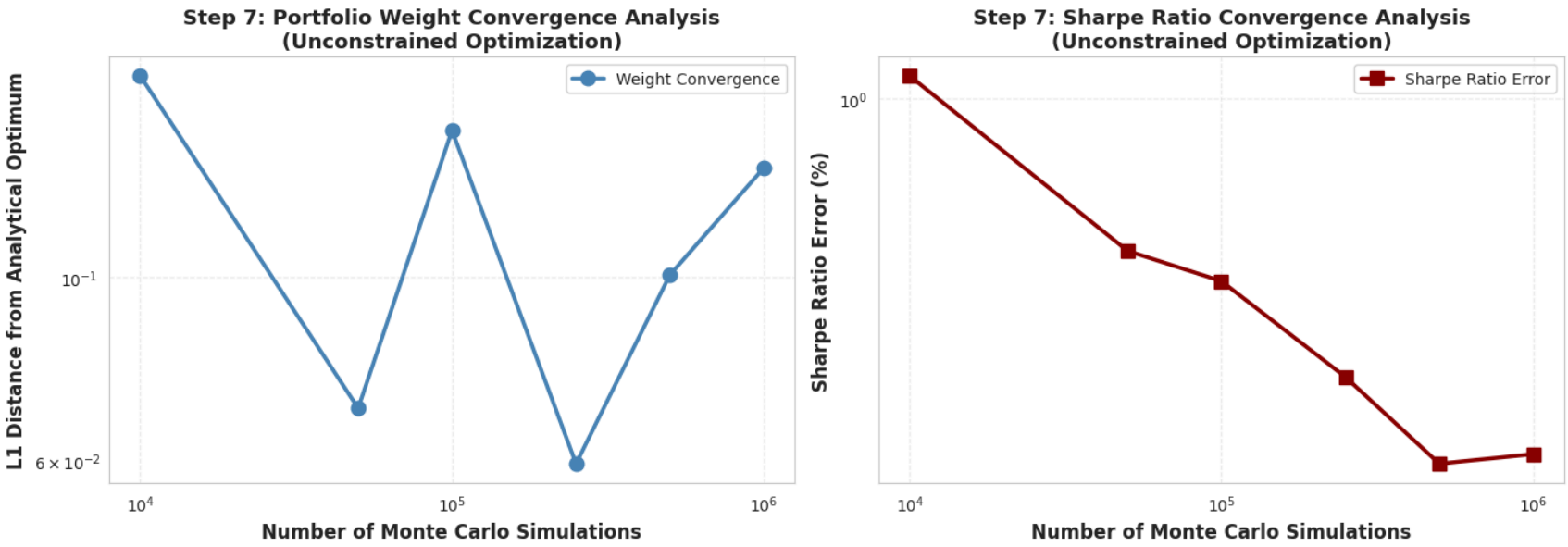
```
subplot_axes_comparison[0].legend(fontsize=11, loc='upper right', framealpha=0.97)
subplot_axes_comparison[0].grid(True, alpha=0.35, linestyle='--', linewidth=1)
subplot_axes_comparison[0].set_xlim(5000, 2000000)

x_position_assets = np.arange(len(asset_universe))
bar_width = 0.38

subplot_axes_comparison[1].bar(x_position_assets - bar_width/2, optimal_weights_step7, bar_width, label='Step
subplot_axes_comparison[1].bar(x_position_assets + bar_width/2, optimal_weights_step8, bar_width, label='Step

subplot_axes_comparison[1].axhline(y=0.30, color='black', linestyle='--', linewidth=2.8, label='Maximum Constr
subplot_axes_comparison[1].set_ylabel('Portfolio Weight Allocation', fontsize=13, fontweight='bold')
subplot_axes_comparison[1].set_xlabel('Asset', fontsize=13, fontweight='bold')
subplot_axes_comparison[1].set_title('Optimal Allocation Comparison\nUnconstrained vs Box-Constrained', fontsi
subplot_axes_comparison[1].set_xticks(x_position_assets)
subplot_axes_comparison[1].set_xticklabels(asset_universe, fontsize=12, fontweight='bold')
subplot_axes_comparison[1].legend(fontsize=11, loc='upper right', framealpha=0.97)
subplot_axes_comparison[1].grid(True, alpha=0.3, axis='y', linestyle='--', linewidth=1)
subplot_axes_comparison[1].set_ylim(0, 0.42)

plt.tight_layout()
plt.show()

print("\n" + "="*80)
print("STEP 8: KEY FINDINGS")
print("="*80)
print(f"\nBox-Constrained Portfolio Optimization Results:")
print(f"  - All weights constrained to [0%, 30%] range")
print(f"  - Optimal Sharpe Ratio (Analytical): {optimal_sharpe_step8:.6f}")
print(f"  - Optimal Annual Return: {optimal_return_step8*100:.2f}%")
print(f"  - Optimal Annual Volatility: {optimal_volatility_step8*100:.2f}%")
print(f"  - Return Change vs Unconstrained: {((optimal_return_step8/optimal_return_step7 - 1)*100):.2f}%")

convergence_threshold_1pct_s8 = None
for result_entry in convergence_results_step8:
    if result_entry['Sharpe_Error_Pct'] <= 1.0:
        convergence_threshold_1pct_s8 = result_entry['Simulations']
        break
```

```
if convergence_threshold_1pct_s8:
    print(f"\nSimulations Required for 1% Sharpe Ratio Error: {convergence_threshold_1pct_s8:,}")
else:
    print(f"\nSimulations Required for 1% Sharpe Ratio Error: Not achieved in tested range")

print(":-) "*5)
```

```
================================================================================
STEP 7: UNCONSTRAINED OPTIMIZATION VS SIMULATION
================================================================================

Asset Universe: ['TSLA', 'WMT', 'BAC', 'LLY', 'AAPL']

Training Period: July 1, 2023 - June 30, 2025

Data Points Available: 520 trading days

Annualized Expected Returns:
  TSLA: 2.118054 (211.81%)
  WMT: -0.808438 (-80.84%)
  BAC: 0.372459 (37.25%)
  LLY: 1.888566 (188.86%)
  AAPL: 0.330331 (33.03%)

Annualized Covariance Matrix:
          TSLA       WMT       BAC       LLY      AAPL
TSLA  6.489085  0.283343  0.719839  1.395570  1.091522
WMT   0.283343  0.501867  0.217307  0.125899  0.200824
BAC   0.719839  0.217307  0.939397  0.361387  0.359580
LLY   1.395570  0.125899  0.361387  3.932990  0.701018
AAPL  1.091522  0.200824  0.359580  0.701018  1.173242

================================================================================
STEP 7: OPTIMIZATION APPROACH
================================================================================

Optimal Portfolio Weights (Optimization):
  TSLA: 0.339745 (33.97%)
  WMT: 0.000000 (0.00%)
  BAC: 0.095501 (9.55%)
  LLY: 0.564753 (56.48%)
  AAPL: 0.000000 (0.00%)

Optimal Portfolio Annual Return: 1.821743 (182.17%)
Optimal Portfolio Annual Volatility: 1.622724 (162.27%)
Optimal Sharpe Ratio: 1.122645

Simulation Count: 10000
  Best Simulated Weights: ['0.356669', '0.009471', '0.007640', '0.581342', '0.044877']
```

```
  Best Simulated Sharpe Ratio: 1.109953
  L1 Distance from Optimal: 0.17572212
  Sharpe Ratio Difference: 0.01269219 (1.1306%)

Simulation Count: 50000
  Best Simulated Weights: ['0.347064', '0.004888', '0.060808', '0.573275', '0.013966']
  Best Simulated Sharpe Ratio: 1.117746
  L1 Distance from Optimal: 0.06938688
  Sharpe Ratio Difference: 0.00489965 (0.4364%)

Simulation Count: 100000
  Best Simulated Weights: ['0.382537', '0.001844', '0.020070', '0.582389', '0.013160']
  Best Simulated Sharpe Ratio: 1.118489
  L1 Distance from Optimal: 0.15086200
  Sharpe Ratio Difference: 0.00415599 (0.3702%)

Simulation Count: 250000
  Best Simulated Weights: ['0.331486', '0.002302', '0.074050', '0.586614', '0.005548']
  Best Simulated Sharpe Ratio: 1.120178
  L1 Distance from Optimal: 0.05942033
  Sharpe Ratio Difference: 0.00246712 (0.2198%)

Simulation Count: 500000
  Best Simulated Weights: ['0.317306', '0.001465', '0.143771', '0.536831', '0.000627']
  Best Simulated Sharpe Ratio: 1.121106
  L1 Distance from Optimal: 0.10072535
  Sharpe Ratio Difference: 0.00153937 (0.1371%)

Simulation Count: 1000000
  Best Simulated Weights: ['0.350348', '0.000166', '0.027587', '0.619963', '0.001937']
  Best Simulated Sharpe Ratio: 1.121024
  L1 Distance from Optimal: 0.13582932
  Sharpe Ratio Difference: 0.00162143 (0.1444%)

--------------------------------------------------------------------------------
STEP 7: CONVERGENCE ANALYSIS
--------------------------------------------------------------------------------
 Sample_Size  L1_Weight_Distance  Sharpe_Error_Percentage
       10000            0.175722                 1.130561
       50000            0.069387                 0.436438
      100000            0.150862                 0.370196
      250000            0.059420                 0.219760
      500000            0.100725                 0.137120
```

```
1000000            0.135829                  0.144429
```




```
================================================================================
STEP 7: KEY FINDINGS
================================================================================

Unconstrained Portfolio Optimization Results:
  - Number of actively invested assets: 3
  - Optimal Sharpe Ratio (Analytical): 1.122645
  - Optimal Annual Return: 182.17%
  - Optimal Annual Volatility: 162.27%

Simulations Required for 1% Sharpe Ratio Error: 50,000
```